

Day 3 Report – Exploratory Data Analysis (EDA)

Project Name

Bluestock Mutual Fund Analytics Capstone

Prepared By

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Objective

The objective of Day 3 was to perform Exploratory Data Analysis (EDA) on the cleaned mutual fund datasets and generate meaningful insights through visualizations and statistical analysis.

Tasks Completed

1. NAV Trend Analysis

- Analyzed NAV trends from 2022 to 2026.
- Created line charts to study fund performance over time.
- Examined monthly average NAV movement.
- Visualized distribution of NAV values.

Charts Created

- NAV Trend Analysis
 - Monthly Average NAV Trend
 - Distribution of NAV Values
 - Top 10 Funds by Latest NAV
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2. AUM Growth Analysis

- Analyzed Assets Under Management (AUM) across fund houses.
- Compared fund houses based on total AUM.
- Identified leading fund houses in the industry.

Charts Created

- AUM by Fund House
 - Top 10 Fund Houses by AUM
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3. SIP Inflow Analysis

- Studied monthly SIP inflow trends.
- Analyzed SIP AUM growth.
- Evaluated active SIP account growth.

- Examined Year-over-Year SIP growth.

Charts Created

- Monthly SIP Inflow Trend
 - SIP AUM Growth
 - Active SIP Accounts Growth
 - SIP YoY Growth %
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4. Category Inflow Analysis

- Performed category-wise inflow analysis.
- Created heatmap visualization to identify investment preferences across categories.

Charts Created

- Category Wise Mutual Fund Inflows Heatmap
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5. Investor Demographic Analysis

- Analyzed investor distribution by age group.
- Studied gender-wise participation.
- Examined SIP contribution patterns by age group.

Charts Created

- Age Group Distribution
 - Gender Distribution
 - SIP Amount by Age Group
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6. Geographic Distribution Analysis

- Analyzed investor participation across states.
- Compared investments between T30 and B30 cities.

Charts Created

- SIP Amount by State
 - T30 vs B30 Distribution
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7. Folio Growth Analysis

- Studied growth in mutual fund folios from 2022 to 2025.
- Marked key milestones in investor participation.

Charts Created

- Mutual Fund Folio Growth
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8. Correlation Analysis

- Calculated daily NAV returns.
- Generated correlation matrix for selected mutual funds.
- Identified relationships among fund performances.

Charts Created

- NAV Return Correlation Matrix
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9. Sector Allocation Analysis

- Aggregated portfolio holdings by sector.
- Evaluated sector-wise exposure across mutual funds.

Charts Created

- Sector Allocation Donut Chart
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Key Findings

1. NAV values showed a strong upward trend during the study period.
2. Most schemes benefited significantly from the 2023 market rally.
3. Temporary corrections were observed during 2024.
4. SBI Mutual Fund maintained the highest AUM among fund houses.
5. SIP inflows reached record levels by the end of 2025.
6. Active SIP accounts increased consistently over time.
7. Investor participation expanded significantly across the industry.
8. Banking and Information Technology sectors dominated portfolio allocations.
9. Tier-1 and Tier-2 cities contributed the majority of investments.

10. Several mutual fund schemes exhibited strong positive return correlations.

Deliverables Completed

- EDA_Analysis.ipynb
 - 15+ Data Visualizations
 - Exported PNG Charts
 - Correlation Matrix
 - Sector Allocation Analysis
 - Investor Demographic Analysis
 - Geographic Analysis
 - Key EDA Findings Documentation
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Conclusion

Day 3 focused on understanding mutual fund industry trends through exploratory data analysis. The analysis revealed strong industry growth driven by increasing SIP participation, rising AUM, expanding investor base, and positive long-term fund performance. The generated visualizations and insights provide a solid foundation for business recommendations and final project reporting.